3. Measurable functions

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Definition 3.1 Suppose that E is a measurable set. We say that a function $f : E \to \mathbb{R}$ is *(Lebesgue-)measurable* if for any interval $I \subseteq \mathbb{R}$,

$$f^{-1}(I) = \{x \in \mathbb{R} : f(x) \in I\} \in \mathcal{M}.$$

Theorem 3.3. *The following conditions are equivalent:*

(i) f *is measurable,*

(ii) for all a*,* $f^{-1}((a, \infty))$ *is measurable,*

(iii) for all a*,* $f^{-1}([a, \infty))$ *is measurable,*

(iv) for all a*,* $f^{-1}((-\infty, a))$ *is measurable,*

(v) for all a*,* $f^{-1}((-\infty, a])$ *is measurable.*

Crux. The preimage mapping f^{-1} strictly preserves all set operations (unions, intersections, complements). Thus, measurability relies entirely on generating the target sets in $\mathbb{R}$.

Equivalence of (ii)-(v): We establish a circular chain of implications using countable operations and complements:

- (ii) $\Rightarrow$ (iii): $[a, \infty) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, \infty\right)$
- (iii) $\Rightarrow$ (iv): $(-\infty, a) = [a, \infty)^c$
- (iv) $\Rightarrow$ (v): $(-\infty, a] = \bigcap_{n=1}^{\infty} \left(-\infty, a + \frac{1}{n}\right)$
- (v) $\Rightarrow$ (ii): $(a, \infty) = (-\infty, a]^c$

Equivalence to (i): Condition (i) trivially implies the others. Conversely, any bounded interval can be constructed via intersection (e.g., $[a, b) = [a, \infty) \cap (-\infty, b)$). Since these intervals generate the Borel σ-algebra $\mathcal{B}$, and f^{-1} preserves σ-algebra generation, conditions (ii)-(v) imply f is measurable. $\square$

Lemma 3.3a. Let $(\Omega, \mathcal{F})$ be a measurable space, and let $(S, \mathcal{H})$ be a measurable space where $\mathcal{H}$ is generated by a collection of sets $\mathcal{E}$ (i.e., $\mathcal{H} = \sigma(\mathcal{E})$). A function $f : \Omega \to S$ is measurable if and only if $f^{-1}(E) \in \mathcal{F}$ for all $E \in \mathcal{E}$.

Crux. The forward direction is trivial by definition. For the reverse direction, assume $f^{-1}(E) \in \mathcal{F}$ for all generators $E \in \mathcal{E}$.

Define the collection of "good sets" in the target space:

$$\mathcal{G} = \{B \subset S : f^{-1}(B) \in \mathcal{F}\}$$

Because the inverse image operation perfectly commutes with countable unions and complements, and because $\mathcal{F}$ is a σ-field, $\mathcal{G}$ itself must be a σ-field.

By our initial assumption, every generator $E \in \mathcal{E}$ has a measurable pre-image, so $\mathcal{E} \subset \mathcal{G}$. Since $\mathcal{G}$ is a σ-field containing $\mathcal{E}$, it must also contain the smallest σ-field generated by $\mathcal{E}$. Therefore, $\mathcal{H} = \sigma(\mathcal{E}) \subset \mathcal{G}$. This implies that for every $B \in \mathcal{H}$, $B \in \mathcal{G}$, which strictly means $f^{-1}(B) \in \mathcal{F}$. $\square$

Note: Since the generator set for $\mathcal{B}(\mathbb{R})$, the Borel σ-algebra, is the set of intervals, we can now assert that if a function $f : \mathbb{R} \to \mathbb{R}$ is measurable according to Definition 3.1, then the inverse image of any Borel set $B \in \mathcal{B}(\mathbb{R})$ in the image space is in $\mathcal{M}$, the σ-field of the domain space. i.e., $f^{-1}(B) \in \mathcal{M},\ \forall B \in \mathcal{B}(\mathbb{R})$.

Certain important measurable functions and sets:

- Constant functions are measurable.
- Continuous functions are measurable.
- The indicator function $\mathbb{I}_A(x)$ is measurable if and only if the set A is measurable.
- Monotone functions are measurable.
- If a function f is measurable, so is the level set $\{x : f(x) = a\}$ for any $a \in \mathbb{R}$.

Theorem 3.5. *The set of real-valued measurable functions defined on $E \in \mathcal{M}$ is a vector space and closed under multiplication, i.e. if f and g are measurable functions then $f + g$, and fg are also measurable (in particular, if g is a constant function $g \equiv c$, cf is measurable for all real c).*

Crux.

- **Addition $(f + g)$:** Because the rationals $\mathbb{Q}$ are dense in $\mathbb{R}$, $f(x) + g(x) < a \iff \exists q \in \mathbb{Q}$ such that $f(x) < q < a - g(x)$. This allows us to decouple the functions into a countable union of measurable intersections:

  $$\{x : f(x) + g(x) < a\} = \bigcup_{q \in \mathbb{Q}} \Big(\{x : f(x) < q\} \cap \{x : g(x) < a - q\} \Big)$$

  Since $\mathbb{Q}$ is countable, this set is measurable.

- **Squares (h^2):** If h is measurable, h^2 is measurable because for any $a \geq 0$:

$$\{x : h^2(x) > a\} = \{x : h(x) > \sqrt{a}\} \cup \{x : h(x) < -\sqrt{a}\}$$

(For $a < 0$, the set is trivially the whole space).

- **Multiplication (fg):** Apply the polarization identity to express the product purely in terms of sums and squares:

$$fg = \frac{1}{4}\left[(f+g)^2 - (f-g)^2\right]$$

Since scalar multiplication, addition, and squaring all preserve measurability, fg is measurable.

$\square$

Lemma 3.7 Suppose that the function $F : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is continuous. If f and g are measurable, then $h(x) = F(f(x), g(x))$ is also measurable.

Crux. Because F is continuous, the preimage $V = F^{-1}((a, \infty))$ is an open set in $\mathbb{R}^2$. Any open set in $\mathbb{R}^2$ can be written as a countable union of open rectangles: $V = \bigcup_{n=1}^{\infty}(a_n, b_n) \times (c_n, d_n)$. Thus, the preimage under the joint mapping $h(x)$ decouples cleanly into measurable 1D components:

$$h^{-1}((a, \infty)) = \{x : (f(x), g(x)) \in V\} = \bigcup_{n=1}^{\infty}\left(f^{-1}((a_n, b_n)) \cap g^{-1}((c_n, d_n))\right)$$

Since f and g are measurable, this countable union of intersections is measurable. $\square$

Proposition 3.8 Let E be a measurable subset of $\mathbb{R}$ and define the non-negative components:

$$f^+(x) \triangleq \begin{cases} f(x) & \text{if } f(x) > 0 \\ 0 & \text{if } f(x) \leq 0 \end{cases}$$

$$f^-(x) \triangleq \begin{cases} 0 & \text{if } f(x) > 0 \\ -f(x) & \text{if } f(x) \leq 0 \end{cases}$$

(i) $f : E \to \mathbb{R}$ is measurable $\iff$ both f^+ and f^- are measurable.

(ii) If f is measurable, then so is $|f|$; but the converse is false.

Crux. (i) **Forward:** Let $A = f^{-1}((0, \infty))$. Since f is measurable, $A \in \mathcal{M}$. Then $f^+ = f \cdot \mathbb{I}_A$ and $f^- = -f \cdot \mathbb{I}_{A^c}$ are products of measurable functions. **Reverse:** $f = f^+ - f^-$, which is a difference of measurable functions.

(ii) **Forward:** $|f| = f^+ + f^-$, which is a sum of measurable functions. **Reverse (Counterexample):** Let V be the non-measurable Vitali set. Define $f = \mathbb{I}_V - \mathbb{I}_{V^c}$. Then $|f| \equiv 1$, which is constant and measurable. However, $f^{-1}((0,\infty)) = V \notin \mathcal{M}$, so f itself is not measurable.

$\square$

Exercise 3.3 If f is measurable, then the truncation of f:

$$f^a(x) \triangleq \begin{cases} a & \text{if } f(x) > a \\ f(x) & \text{if } f(x) \le a \end{cases}$$

is also measurable.

Theorem 3.9. *If $\{f_n\}$ is a sequence of measurable functions defined on the set E in $\mathbb{R}$, then the following are measurable functions also:*

$$\max_{n\le k} f_n,\ \min_{n\le k} f_n,\ \sup_{n\in\mathbb{N}} f_n,\ \inf_{n\in\mathbb{N}} f_n,\ \limsup_{n\to\infty} f_n,\ \liminf_{n\to\infty} f_n$$

Crux. Measurability is preserved because extrema translate directly to countable set operations.

- **Supremum & Max:** The supremum exceeds a iff at least one function exceeds a:

$$\{x : \sup_n f_n(x) > a\} = \bigcup_{n=1}^{\infty} \{x : f_n(x) > a\}$$

  (Finite union for max).

- **Infimum & Min:** The infimum is at least a iff every function is at least a:

$$\{x : \inf_n f_n(x) \ge a\} = \bigcap_{n=1}^{\infty} \{x : f_n(x) \ge a\}$$

  (Finite intersection for min). **Note:** We use $\ge$ here rather than $>$. If we used $>$, the intersection logic fails (e.g., if $f_n(x) = 1/n$, then $f_n(x) > 0$ for all n, but $\inf f_n(x) = 0 \not> 0$).

- **Limsup & Liminf:** These are defined by applying countable supremum and infimum operations sequentially:

$$\limsup_{n\to\infty} f_n = \inf_{k\ge 1}\left(\sup_{n\ge k} f_n\right)$$
$$\liminf_{n\to\infty} f_n = \sup_{k\ge 1}\left(\inf_{n\ge k} f_n\right)$$

Since applying a countable sup to $\{f_n\}$ yields a new sequence of measurable functions, subsequently applying a countable inf to that sequence also yields a measurable function. **Note:** Composition of measurable functions *is not* necessarily a measurable function, so we must rely on these sequential limits rather than functional composition.

□

Corollary 3.10 If a sequence $\{f_n\}$ of measurable functions converges pointwise, then the limit $f = \lim_{n\to\infty} f_n$ is a measurable function.

Crux. Pointwise convergence implies $\lim_{n\to\infty} f_n = \limsup_{n\to\infty} f_n = \liminf_{n\to\infty} f_n$. By Theorem 3.9, both the lim sup and lim inf of a measurable sequence are measurable functions. □

Theorem 3.12. *If $f : E \to \mathbb{R}$ is measurable, $E \in \mathcal{M}$, and $g : E \to \mathbb{R}$ is such that the set $\{x : f(x) \neq g(x)\}$ is null, then g is measurable.*

Crux. Define the difference $d = g - f$. Let $N = \{x : d(x) \neq 0\}$, which is given as a null set ($m(N) = 0$). By the completeness of the Lebesgue measure, any subset of N is automatically measurable.

- For $a \geq 0$: $d^{-1}((a, \infty)) = \{x : d(x) > a\} \subseteq N$. Hence, it is null and measurable.
- For $a < 0$: $d^{-1}((-\infty, a]) = \{x : d(x) \leq a\} \subseteq N$. Hence, it is null and measurable. Its complement, $d^{-1}((a, \infty))$, must therefore also be measurable.

Thus, $d^{-1}((a, \infty))$ is measurable for all $a \in \mathbb{R}$, proving d is a measurable function. Since f is measurable, $g = f + d$ is also measurable. □

Corollary 3.13 If $\{f_n\}$ is a sequence of measurable functions and $f_n \to f$ almost everywhere on E, then f is measurable.

Crux. Let $N = \{x \in E : f_n(x) \not\to f(x)\}$. By definition of almost everywhere convergence, N is a null set.

- **Everywhere Convergence:** The truncated sequence $f_n \mathbb{I}_{N^c}$ converges *everywhere* to $f\mathbb{I}_{N^c}$. Since each $f_n \mathbb{I}_{N^c}$ is measurable, their pointwise limit $f\mathbb{I}_{N^c}$ is measurable (by Corollary 3.10).
- **Almost Everywhere Equivalence:** The functions $f\mathbb{I}_{N^c}$ and f differ only on the null set N. Since $f\mathbb{I}_{N^c}$ is measurable, f is also measurable (by Theorem 3.12).

□

Exercise 3.5 Let $\{f_n\}$ be a sequence of measurable functions. The set $E = \{x : f_n(x) \text{ converges}\}$ is measurable.

Crux. A sequence converges to a finite limit if and only if its limit superior and limit inferior are equal and finite. Let $h(x) = \limsup_{n\to\infty} f_n(x)$ and $g(x) = \liminf_{n\to\infty} f_n(x)$. Both are measurable (Theorem 3.9). The set of points where the sequence converges is exactly:

$$E = \{x : h(x) - g(x) = 0\} \cap \{x : h(x) < \infty\} \cap \{x : g(x) > -\infty\}$$

Since h and g are measurable, their difference $h - g$ is measurable wherever it is finite. Thus, E is the intersection of strictly measurable sets and is therefore measurable. $\square$

Definition 3.14 Suppose $f : E \to \overline{\mathbb{R}}$ is measurable. The *essential supremum* $\operatorname{ess\,sup} f$ is defined as $\inf\{z : f \le z \text{ a.e.}\}$ and the *essential infimum* $\operatorname{ess\,inf} f$ is $\sup\{z : f \ge z \text{ a.e.}\}$.

Lemma $f \le \operatorname{ess\,sup} f$ a.e.

Crux. Let $S = \operatorname{ess\,sup} f$. If $S = \infty$, the result is trivial. If $S < \infty$, then for any $n \in \mathbb{N}$, the value $S + 1/n$ is strictly greater than the infimum of all essential upper bounds, making it an essential upper bound itself. Thus, the set $A_n = \left\{x : f(x) > S + \frac{1}{n}\right\}$ is a null set. The set of points where f strictly exceeds its essential supremum can be written as:

$$\{x : f(x) > S\} = \bigcup_{n=1}^{\infty} \left\{x : f(x) > S + \frac{1}{n}\right\} = \bigcup_{n=1}^{\infty} A_n$$

Since a countable union of null sets is null, $\{x : f(x) > S\}$ is a null set, proving $f \le S$ a.e. $\square$

Proposition 3.15 If f, g are measurable functions, then

$$\operatorname{ess\,sup}(f+g) \le \operatorname{ess\,sup} f + \operatorname{ess\,sup} g.$$

Crux. Let $S_f = \operatorname{ess\,sup} f$ and $S_g = \operatorname{ess\,sup} g$. By the preceding lemma, the exceedance sets $A = \{x : f(x) > S_f\}$ and $B = \{x : g(x) > S_g\}$ are null.

For the sum to exceed $S_f + S_g$, at least one function must exceed its respective essential supremum. Thus:

$$\{x : f(x) + g(x) > S_f + S_g\} \subseteq A \cup B$$

Since $A \cup B$ is a finite union of null sets, it is also null. Therefore, $f + g \le S_f + S_g$ a.e., which immediately implies $\operatorname{ess\,sup}(f+g) \le S_f + S_g$. $\square$

Exercise 3.6 For measurable f, $\operatorname{ess\,sup} f \le \sup f$. These quantities coincide when f is continuous.

Crux. Let $S_{ess} = \operatorname{ess\,sup} f$ and $S_{sup} = \sup f$.

Inequality: By definition, $f(x) \leq S_{sup}$ everywhere, which means it holds almost everywhere. Thus, S_{sup} is an valid essential upper bound. Since S_{ess} is the infimum of all essential upper bounds, $S_{ess} \leq S_{sup}$.

Equality for continuous f: Assume for contradiction that $S_{ess} < S_{sup}$. By definition of the supremum, there must exist points where $f(x) > S_{ess}$. Therefore, the preimage $V = f^{-1}((S_{ess}, \infty))$ is non-empty. Because f is continuous, V is an open set. In $\mathbb{R}$, every non-empty open set has strictly positive Lebesgue measure ($m(V) > 0$). However, V is exactly the set $\{x : f(x) > S_{ess}\}$, which must be a null set by the definition of the essential supremum. This is a contradiction. Thus, $S_{ess} = S_{sup}$. $\square$

Definition: Random Variables. If $(\Omega, \mathcal{F}, P)$ is a probability space, then $X : \Omega \to \mathbb{R}$ is a *random variable* if $\forall a \in \mathbb{R}$, the set $X^{-1}([a, \infty)) \in \mathcal{F}$:

$$\{\omega \in \Omega : X(\omega) \geq a\} \in \mathcal{F}.$$

Definition: σ-field generated by a random variable: If X is a random variable in the probability space $(\Omega, \mathcal{F}, P)$, then the set

$$\mathcal{F}_X \triangleq X^{-1}(\mathcal{B}) = \{S \in \mathcal{F} : S = X^{-1}(B) \text{ for some } B \in \mathcal{B}\}$$

is a σ-field that is a subset of $\mathcal{F}$ and is called the σ-field *generated* by X.

Exercise 3.7 The σ-algebra $\mathcal{F}_X$ is the smallest σ-algebra containing the inverse images $X^{-1}(B)$ of all Borel sets B.

Crux. By definition, $\mathcal{F}_X = \{X^{-1}(B) : B \in \mathcal{B}\}$. If $\mathcal{G}$ is any σ-algebra containing $X^{-1}(B)$ for all $B \in \mathcal{B}$, then $\mathcal{G}$ must contain the entire collection of these sets. Therefore, $\mathcal{F}_X \subseteq \mathcal{G}$, proving $\mathcal{F}_X$ is the minimal σ-algebra containing these preimages. $\square$

Definition: Probability Distribution. If X is a random variable in the probability space $(\Omega, \mathcal{F}, P)$, then the measure

$$P_X(B) = P(X^{-1}(B))$$

defined on the σ-field of Borel sets $\mathcal{B}$ is called the *probability distribution* of the random variable X. Lemma 3.3a assures us that $X^{-1}(B) \in \mathcal{F}$ so that this definition is perfectly valid.

Theorem 3.16. *The set function P_X is countably additive.*

Crux. Let $\{B_i\}$ be a sequence of pairwise disjoint Borel sets. Because the preimage operation preserves set disjointness, the sets $X^{-1}(B_i)$ are also pairwise disjoint in the underlying sample space.

Applying the definition of the induced measure P_X and the countable additivity of the underlying probability measure P:

$$\begin{aligned} P_X\left(\bigcup_i B_i\right) &= P\left(X^{-1}\left(\bigcup_i B_i\right)\right) \\ &= P\left(\bigcup_i X^{-1}(B_i)\right) \\ &= \sum_i P(X^{-1}(B_i)) \\ &= \sum_i P_X(B_i) \end{aligned}$$

□

Note: With the theorem above along with the fact that $P_X(\mathbb{R}) = P(\Omega) = 1$, we can conclude that $(\mathbb{R}, \mathcal{B}, P_X)$ is a probability space.

Definition 3.18: Independent Random Variables. X and Y are *independent* if the σ-fields generated by them are independent. In other words, for any Borel sets $B, C \in \mathcal{B}$,

$$P(X^{-1}(B) \cap Y^{-1}(C)) = P(X^{-1}(B))P(Y^{-1}(C)).$$